

Yen-Hsun LIN

Curriculum Vitae

Institute of Physics, Academia Sinica
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RESEARCH SUMMARY

I am an astroparticle physicist with expertise in multimessenger astronomy and dark matter (DM) detection. My research focuses on *three key areas*: (1) supernova-neutrino-boosted DM, (2) anomalous heating from DM in compact stars, and (3) probing DM self-interactions and DM-nucleon interactions in stars and planets. The first area is particularly vital as it opens the *new possibility* for direct DM mass measurements using *time-of-flight techniques*. I also collaborate with DUNE/COHERENT members and work on reducing systematic uncertainties in DUNE-like detectors. Additionally, I contributed to the JUNO collaboration, assessing its data analysis to solar-captured DM. My background in astroparticle physics and extensive research experience have provided me with a deep understanding of DM and its broader implications to our Universe.

Topic of Interest

Astroparticle physics, dark matter physics, supernova and compact star physics, high performance computation, Bayesian inference, and Monte Carlo simulation.

Programming

Python, Cython, C++, Mathematica and Matlab.

EDUCATION

National Chiao Tung University

PhD of the Institute of Physics

Thesis: Indirect detection of dark matter through neutrinos

Advisor: Prof. Guey-Lin Lin

Hsinchu, Taiwan
Aug. 2011 – Jun. 2016

National Chiao Tung University

Master of the Institute of Physics (direct to PhD program)

Advisor: Prof. Guey-Lin Lin

Hsinchu, Taiwan
Aug. 2010 – Jul. 2011

National Chiao Tung University

Bachelor of the Department of Electrophysics

Hsinchu, Taiwan
Aug. 2006 – Jul. 2010

EXPERIENCE

Postdoctoral Scholar

Institute of Physics, Academia Sinica

Host: Dr. Meng-Ru Wu

Taipei, Taiwan
Aug. 2023 – Present

Visiting Scholar

School of Physics, Melbourne University

Host: Prof. Nicole F. Bell

Melbourne, Australia
Oct. 2023 – Nov. 2023

Postdoctoral Scholar

Physics Division, National Center for Theoretical Sciences

Taipei, Taiwan
Dec. 2021 – Jul. 2023

Distinguished Postdoctoral Scholar

Institute of Physics, Academia Sinica

Host: Dr. Meng-Ru Wu

Taipei, Taiwan
Aug. 2019 – Dec. 2021

Postdoctoral Researcher

Department of Physics, National Cheng Kung University

Host: Prof. Chuan-Hung Chen

Tainan, Taiwan
Oct. 2017 – Jul. 2019

HONORS & AWARDS

1. **Postdoctoral Academic Research Award** Taiwan, 2024
Awarded by the Division of Particles and Fields, Taiwan Physical Society.
2. **NCTS Postdoc Paper Award** Taiwan, 2024
Awarded by the Physics Division, National Center for Theoretical Sciences (NCTS).
3. **Best Research Paper Award for Junior Research Investigator**
Awarded by the Institute of Physics, Academia Sinica. Taiwan, 2024
4. **Selected Participant of the 13th HOPE Meeting with Nobel Laureates**
Representative of Taiwan. Japan, 2022
5. **Distinguished Postdoctoral Scholar** Taiwan, 2019
Independent position with grant, selected by the Academia Sinica.
6. **Postgraduate Student Thesis Award** Taiwan, 2017
Selected as the best PhD thesis of the year in physical sciences, awarded by the Taiwan Physical Society.
7. **Selected Honorary Member of the Phi Tau Phi Scholastic Society**
Issued to the student graduated with top score. Taiwan, 2016

COLLABORATION MEMBERSHIP

1. With Members of DUNE/COHERENT Collaborations USA
2020 – 2024
 - ◇ Collaborating with Dr. Gianluca Petrillo and Dr. Yun-Tse Tsai
 - ◇ Analysis the impact due to ν_e -Ar cross section uncertainty
 - ◇ Improving pinched parameter sensitivity via Machine Learning

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|--|-----------------|
| 2. Jiangmen Underground Neutrino Observatory (JUNO) | Jiangmen, China |
| ◇ Co-author of the JUNO Yellow Book (R&D tech notes) | 2015 – 2016 |
| ◇ Sensitivity projection for the solar-captured DM in JUNO | |

ADVISEES

PhD students

- | | |
|--|-----------------------|
| 1. Vo Quang Nhat | Hsinchu, Taiwan |
| <i>Institute of Physics, NYCU</i> | Aug. 2022 – Jul. 2023 |
| Co-supervising with Prof. Guey-Lin Lin | |
| 2. Lam Thi To Uyen | Hsinchu, Taiwan |
| <i>Institute of Physics, NYCU</i> | Aug. 2022 – Jul. 2023 |
| Co-supervising with Prof. Guey-Lin Lin | |

Undergraduates

- | | |
|---|-----------------------|
| 1. Tsung-Han Tsai (ASIoP Summer Student Program) | Hsinchu, Taiwan |
| <i>Department of Physics, NTHU</i> | Jul. 2022 – Aug. 2022 |
| Co-supervising with Dr. Meng-Ru Wu and work published in <i>Phys. Rev. D</i> 108 , 083013 (2023). | |
| Tsung-Han has received an offer for enrollment in the PhD program at UW Madison (US). | |
| 2. Yong Sheng Yap (ASIoP Summer Student Program) | Hsinchu, Taiwan |
| <i>Department of Physics, NTHU</i> | Jul. 2021 – Aug. 2021 |
| Co-supervising with Dr. Meng-Ru Wu and currently a PhD student in Cambridge University (UK). | |
| 3. Wen-Hua Wu (ASIoP Summer Student Program) | Taipei, Taiwan |
| <i>Department of Physics, NTU</i> | Jul. 2020 – Aug. 2020 |
| Co-supervising with Dr. Meng-Ru Wu and work published in <i>Phys. Rev. Lett.</i> 130 , 111002 (2023). Currently a PhD student in Rice University (US). | |
| 4. Adeela Malik (SLAC Summer Student Program) | San Antonio, USA |
| <i>Department of Physics, University of Texas at San Antonio</i> | Jul. 2020 – Aug. 2020 |
| Co-supervising with and Prof. Hirohisa Tanaka (SLAC) and Dr. Yun-Tse Tsai (SLAC). | |

GITHUB REPOSITORIES

- **snorer: *S*pernova-*N*eutrino-*b*Oosted *daRk* *maTT**ER***
Description: Evaluating the time-of-flight signatures of boosted dark matter due to supernova neutrinos from Milky Way, SN1987a and arbitrary distant galaxy.
Role: Main developer and maintainer
Project Page: <https://github.com/yenhhsunlin/snorer>
- **dukes: *Diff*Use-boosted *darK* *maTT**Er* by *S*upernova neutrinos**
Description: Evaluating the signatures of diffuse boosted dark matter by supernova neutrinos in the early Universe.

Role: Main developer and maintainer

Project Page: <https://github.com/yenhsunlin/dukes>

- dynesor: *DY*namical *NE*sted *S*ampling *integratOR*

Description: MCMC integrator for evaluating multidimensional integration based on dynamical nested sampling.

Role: Main developer and maintainer

Project Page: Non-disclose.

SCIENTIFIC ACTIVITIES & SERVICES

Workshop organization

- Organizer of the Mini-workshop on Novel Experimental and Astrophysical Probes for Dark Matter, Taipei, Taiwan, 2021

Journal referee

1. Physics Letters B
2. Annals of Physics

Publication List

REFEREED ARTICLES

Dagger (†) and asterisk (*) indicate *first author* and *corresponding author*, respectively. Otherwise, the author list is arranged alphabetically.

1. **Y.-H. Lin**^{†,*} and M.-R. Wu, *Mapping the evolution of supernova-neutrino-boosted dark matter within the Milky Way*, *Phys. Rev. D* **112**, 083001 (2025) [[arXiv:2506.15151](#)]
2. **Y.-H. Lin**^{†,*} and M.-R. Wu, *Supernova-neutrino-boosted dark matter from all galaxies*, *Phys. Rev. Lett.* **133**, 111004 (2024) [[arXiv:2404.08528](#)]
3. **Y.-H. Lin**^{†,*}, T.-H. Tsai, G.-L. Lin, H. T.-K. Wong and M.-R. Wu, *Signatures of afterglows from light dark matter boosted by supernova neutrinos in current and future large underground detectors*, *Phys. Rev. D* **108**, 083013 (2023) [[arXiv:2307.03522](#)]
4. **Y.-H. Lin**^{†,*}, W.-H. Wu, M.-R. Wu and H. T.-K. Wong, *Searching for afterglow: Light dark matter boosted by supernova neutrinos*, *Phys. Rev. Lett* **130**, 111002 (2023) [[arXiv:2206.06864](#)]
5. A. Bauswein, G. Guo, J.-H. Lien, **Y.-H. Lin** and M.-R. Wu, *Compact dark objects in neutron star mergers*, *Phys. Rev. D* **107**, 083002 (2023) [[arXiv:2012.11908](#)]
6. G.-L. Lin and **Y.-H. Lin**^{*}, *Exploring dark sector parameters in light of neutron star temperatures*, *Phys. Rev. D* **104**, 063021 (2021) [[arXiv:2102.11151](#)]
7. G.-L. Lin and **Y.-H. Lin**^{*}, *Analysis on the black hole formations inside old neutron stars by isospin-violating dark matter with self-interaction*, *JCAP* **08**, 022 (2020) [[arXiv:2004.05312](#)]
8. C.-S. Chen and **Y.-H. Lin**^{*}, *Reheating neutron stars with the annihilation of self-interacting dark matter*, *JHEP* **08**, 069 (2018) [[arXiv:1804.03409](#)]
9. C.-S. Chen and **Y.-H. Lin**^{*}, *On the evolution process of two-component dark matter in the Sun*, *JHEP* **04**, 074 (2018) [[arXiv:1802.06956](#)]
10. C.-H. Chen and **Y.-H. Lin**, *Study of $B_c^\pm \rightarrow (D^0 K^\pm, D^0 \pi^\pm)$ decays*, [arXiv:1710.05531](#)
11. C.-S. Chen, G.-L. Lin, **Y.-H. Lin** and F. Xu, *The 17 MeV anomaly in beryllium decays and $U(1)$ portal to dark matter*, *Int. J. Mod. Phys. A* **32**, 1750178 (2017) [[arXiv:1609.07198](#)]
12. C.-S. Chen, G.-L. Lin and **Y.-H. Lin**, *Thermal transport of the solar captured dark matter and its impact on the indirect dark matter search*, *Phys. Dark Univ.* **14**, 35 (2016) [[arXiv:1508.05263](#)]

13. C.-S. Chen, G.-L. Lin and **Y.-H. Lin**, *Complementary test of the dark matter self-interaction by direct and indirect detections*, *JCAP* **01**, 013 (2016) [[arXiv:1505.03781](#)]
14. G.-L. Lin, **Y.-H. Lin** and F.-F. Lee, *Probing the coupling of heavy dark matter to nucleons by detecting neutrino signature from the Earth core*, *Phys. Rev. D* **91**, 033002 (2015) [[arXiv:1409.3094](#)]
15. C.-S. Chen, F.-F. Lee, G.-L. Lin and **Y.-H. Lin**, *Probing dark matter self-interaction in the Sun with IceCube-PINGU*, *JCAP* **10**, 049 (2014) [[arXiv:1408.5471](#)]

COLLABORATION PAPERS

1. A. Abusleme *et al.* [JUNO Collaboration], *First measurement of reactor neutrino oscillations at JUNO*, (2025), [[arXiv:2511.14593](#)]
2. A. Abusleme *et al.* [JUNO Collaboration], *Initial performance results of the JUNO detector*, (2025), [[arXiv:2511.14590](#)]
3. A. Abusleme *et al.* [JUNO & Daya Bay Collaborations], *Optimization of the JUNO liquid scintillator composition using a Daya Bay antineutrino detector*, *Nucl. Instrum. Meth. A* **988**, 164823 (2021) [[arXiv:2007.00314](#)]
4. Z. Djurcic *et al.* [JUNO Collaboration], *JUNO conceptual design report*, (2015), [[arXiv:1508.07166](#)]
5. F. An *et al.* [JUNO Collaboration], *Neutrino Physics with JUNO*, *J. Phys. G* **43**, 1 (2016) [[arXiv:1507.05613](#)]

CONFERENCE PROCEEDINGS

1. G.-L. Lin and **Y.-H. Lin**, *Exploring dark sector parameters in light of neutron star temperatures*, *PoS ICHEP2022*, 106 (2022)
2. Lam T. T. Uyen, G.-L. Lin and **Y.-H. Lin**, *Constraints on lepton-flavor-violating scalar portal using the Belle II result in the search for $e^+e^- \rightarrow e^\pm + \text{invisible}$ with $L = 276 \text{ pb}^{-1}$* , *PoS ICHEP2022*, 1229 (2022)
3. **Y.-H. Lin** and G.-L. Lin, *Analysis on the black hole formations inside old neutron stars by isospin-violating dark matter with self-interaction*, *PoS ICHEP2020*, 598 (2020)
4. **Y.-H. Lin** and G.-L. Lin, *Probing self-interacting dark matter through neutron stars*, *PoS EPS-HEP2019*, 075 (2020)
5. C.-S. Chen, G.-L. Lin **Y.-H. Lin** and F. Xu, *The 17 MeV anomaly in beryllium decays and $U(1)$ portal to dark matter*, *PoS EPS-HEP2017*, 627 (2017)
6. C.-S. Chen, **Y.-H. Lin** and G.-L. Lin, *Complementary test of the dark matter self-interaction in dark $U(1)$ model by direct and indirect detection*, *JPS Conf. Proc.* **14**, 020103 (2017)
7. **Y.-H. Lin**, C.-S. Chen and G.-L. Lin, *Thermal transport of the solar captured dark matter and its implication*, *JPS Conf. Proc.* **14**, 020110 (2017)

8. C.-S. Chen, G.-L. Lin and **Y.-H. Lin**, *Probing dark matter self-interaction in the Sun with IceCube-PINGU*, *PoS FPC2015*, 065 (2015)
9. **Y.-H. Lin**, C.-S. Chen and G.-L. Lin, *Thermal transport of the solar captured dark matter and its implication*, *PoS EPS-HEP2015*, 385 (2015)
10. C.-S. Chen, F.-F. Lee, G.-L. Lin and **Y.-H. Lin**, *The dark matter self-interaction and its impact on the critical mass for dark matter evaporations inside the Sun*, *Nucl. Part. Phys. Proc.* **273**, 347 (2016)

Presentations

INVITED CONFERENCE/WORKSHOP TALKS

1. *Supernova-neutrino-boosted dark matter: From the next galactic SN to all SNe in the Universe*
The 4th Workshop on Symmetry and Structure of the Universe, Jeonju, Korea (2025/8)
2. *Exploring light dark matter boosted by supernova neutrinos in the past and present Universe*
Particle Physics Phenomenology Workshop (PPP 15), Taipei, Taiwan (2024/10)
3. *Detecting afterglow signatures from light dark matter boosted by supernova neutrinos*
Interstellar and Intergalactic Insights: Exploring the Energetic Universe with Multi-messengers, Taipei, Taiwan (2023/12)
4. *Detection of $SN\nu$ BDM in current and future large underground detectors*
The 3rd International Joint Workshop on the Standard Model and Beyond and the 11th KIAS Workshop on Particle Physics and Cosmology, Jeju, Republic of Korea (2023/11)
5. *Searching for afterglow: Light dark matter boosted by supernova neutrinos*
Interplay of Nuclear, Neutrino and BSM Physics at Low-Energies (INT 23-85w), Seattle, USA (2023/4)
6. *Light DM constraints from neutron stars and supernova neutrinos*
Theory Meets Experiment: Particle Physics and Cosmology, Quy Nhon, Vietnam (2023/1)
7. *Light DM constraints from neutron stars and supernova neutrinos*
NCTS Annual Theory Meeting, Taipei, Taiwan (2022/12)
8. *Light dark matter boosted by supernova neutrinos*
Dark Matter in Compact Objects, Stars, and in Low Energy Experiments (INT 22-2b), Seattle, USA (2022/8)
9. *Searching the afterglow from supernova neutrino boosted dark matter*
Particle Physics Phenomenology Workshop (PPP 14), Taipei, Taiwan (2022/6)
10. *Exploring dark matter with compact stars*
Mini-workshop on Novel Experimental and Astrophysical Probes for Dark Matter, Taipei, Taiwan (2021/3)
11. *Probing the isospin violation of self-interacting dark matter through old neutron stars*
NCTS Annual Theory Meeting, Taipei, Taiwan (2019/12)
12. *Probing self-interacting dark matter through neutron stars*
Particle Physics Phenomenology Workshop (PPP 13), Taipei, Taiwan (2019/6)

CONTRIBUTED CONFERENCE/WORKSHOP TALKS

1. *Shimmering darkness: Mapping the distribution of supernova-neutrino-boosted dark matter across the sky*
Neutrino and Dark Matter 2025 (NDM 2025), Paris, France (2025/5)
2. *Exploring light dark matter boosted by supernova neutrinos in the present and past Universe*
International Joint Workshop on the Standard Model and Beyond & the 3rd Gordon Godfrey Workshop on Astroparticle Physics, Sydney, Australia (2024/12)
3. *Searching light dark matter boosted by supernova neutrinos in Super-K, Hyper-K and DUNE*
International Conference on Topics in Astroparticle and Underground Physics (TAUP 2023), Vienna, Austria (2023/8)
4. *Detection of afterglows from supernova-neutrino boosted dark matter in large underground detectors*
International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology, Nantou, Taiwan (2023/8)
5. *Analysis on the black hole formations inside old neutron stars by isospin-violating dark matter with self-interaction*
International Conference on High Energy Physics (ICHEP 2020), Prague, Czech Republic (2020/8)
6. *The art of inference: Practicing Bayesian reasoning in computer vision problems*
PyCon TW, Taipei, Taiwan (2019/9)
7. *Probing self-interacting dark matter through neutron stars*
European Physical Society Conference on High Energy Physics (EPS-HEP 2019), Ghent, Belgium (2019/7)
8. *On the evolution process of two-component dark matter in the Sun*
International Conference on Neutrino Physics and Astrophysics (Neutrino 2018), Heidelberg, Germany (2018/6)
9. *Thermal transport of the solar captured dark matter and its implication*
International Symposium on Nuclei in the Cosmos (NIC-XIV), Niigata, Japan (2016/6)
10. *Thermal transport of the solar captured dark matter and its implication*
European Physical Society Conference on High Energy Physics (EPS-HEP 2015), Vienna, Austria (2015/7)
11. *The dark matter self-interaction and its impact on the critical mass for dark matter evaporations inside the Sun*
International Conference on High Energy Physics (ICHEP 2014), València, Spain (2014/7)
12. *Probing the coupling of heavy dark matter to nucleons by detecting neutrino signature from the Earth core*
International Symposium on Particles, Strings and Cosmology (PASCOS 2014), Taipei, Taiwan (2013/12)

13. *Probing the coupling of heavy dark matter to nucleons by detecting neutrino signature from the Earth core*
International Symposium on Cosmology and Particle Astrophysics (CosPA 2013), Honolulu, Hawaii (2013/6)

SEMINARS & COLLOQUIA

1. *Behind the veil of darkness: A journey into the Uncharted Universe*
Colloquium for the Department of Physics, Tunghai University, Taichung, Taiwan (2024/5)
2. *Exploring light dark matter boosted by supernova neutrinos in the present and past Universe*
Seminar for the Department of Physics, National Tsing Hua University, Hsinchu, Taiwan (2024/5)
3. *Detecting afterglow signatures from light dark matter boosted by supernova neutrinos*
Seminar for the Department of Physics, Chinese University of Hong Kong, Hong Kong (2024/1)
4. *Signatures of afterglows from light dark matter boosted by supernova neutrinos in large underground detectors*
Seminar for the Theoretical Particle Physics (TPP) Group, School of Physics, Melbourne University, Melbourne, Australia (2023/10)
5. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Seminar for the Department of Physics, National Tsing Hua University, Hsinchu, Taiwan (2023/5)
6. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Seminar for the Department of Physics, Chung Yuan Christian University, Taoyuan, Taiwan (2023/3)
7. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Webinar for the Supernova Early Warning System (SNEWS), USA (2023/2)
8. *Search for afterglow: Light dark matter boosted by supernova neutrino*
Seminar for the Department of Physics, National Taiwan Normal University, Taipei, Taiwan (2022/11)
9. *Searching for afterglow: Light dark matter boosted by supernova neutrinos*
NCTS Particle Physics Journal Club, National Taiwan University, Taipei, Taiwan (2022/9)
10. *Searching light to heavy dark matter by supernova neutrinos and neutron star temperature*
Webinar for the Sydney Consortium for Particle Physics and Cosmology (Sydney-CPPC), Sydney, Australia (2021/10)
11. *Probing dark matter with neutron star*
Seminar for the Department of Physics, Chung Yuan Christian University, Taoyuan, Taiwan (2020/11)

12. *Probing dark matter with neutron star*
Seminar for the Center of Astrophysics and Gravity, National Taiwan Normal University, Taipei, Taiwan (2020/7)
13. *Neutron star sensitivity on isospin-violating dark matter with self-interaction*
Seminar for the Department of Physics, National Central University, Taoyuan, Taiwan (2020/6)
14. *Uncharted Universe*
Colloquium for the Department of Physics, Tamkang University, New Taipei, Taiwan (2018/12)
15. *Reheating neutron stars with the annihilation of self-interacting dark matter.*
Seminar for the Institute of Physics, National Chiao Tung University, Hsinchu, Taiwan (2018/5)
16. *Reheating neutron stars with the annihilation of self-interacting dark matter.*
Seminar for the Institute of Physics, Academia Sinica, Taipei, Taiwan (2018/4)
17. *Indirect detection of dark matter through neutrinos*
Seminar for the Department of Physics, Chung Yuan Christian University, Taoyuan, Taiwan (2017/12)